

Manual

Cavity-Related Inspection Planning for FEP FEP-NES 8.1

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1 Cavity-Related Inspection Planning for FEP

Purpose

This component allows for activating cavity-related data collection in inspection planning with a cavity reference to the sample. This allows for the cavity-related collection of data in the inspection process and their subsequent evaluation.

Implementation notes

You use this component if you want to collect and evaluate data in relation to cavities.

Integration

This component is the basis for "Collection/information functions for quality data" using the data collection functions of the AIP data collection client.

Features

The following functions are available:

- Definition of sample-related cavity assignment on the inspection plan level
- Activation / deactivation of inspection plan characteristics for cavity-related data collection

2 Inspection Planning

Overview

Menu	Quality management → In-production inspection → Inspection planning Quality management → Goods receipt → Inspection planning Quality management → Goods issue → Inspection planning Quality management → Initial sample → Inspection planning Quality management → Test equipment management → Inspection planning
Transaction code	iplp ¹
Function authorization	iplp

Usage

According to the area of use, inspection planning is the most important function as the inspection plans are the basis for the creation of inspection orders with subsequent inspections. This refers to inspections in goods receipt, production and goods issue. In addition, there are initial sample inspections and the calibration of test equipment. Since the requirements of inspection planning are almost identical in all areas, the functions of inspection planning in HYDRA are almost identical for these areas.

When creating inspection plans, the user is generally supported by the master data catalogs. For this reason, careful maintenance of master data is very important.



¹If you start the "inspection plan" application by entering the transaction code, you cannot create a new inspection plan.

Integration

Inspection planning is the prerequisite for creating inspection requirements, the related inspection steps and the resulting inspections/calibrations.

The inspection plan used is referenced in the inspection requirements created. In addition, the inspection planning is referenced in some evaluations. As a consequence, the basis of an inspection order characteristic control chart can be extended to the data of the same inspection plan number. The same applies to the global control chart analysis. Here it is possible to use specific inspection plan filter fields. Global control chart evaluation does not take account of any calibration data and hence no filtering according to calibration inspection plans.

Another significant relationship exists with regard to the production control plan. A production control plan cannot be prepared without the prior creation of inspection plans.

Prerequisite

The creation of inspection plans requires the maintenance of the relevant master data. The master data to be maintained depend on the relevant application. However, the prior maintenance of articles and characteristics is essential. As far as the creation of article group inspection plans is concerned, the entry of article groups is also required.

Selection criteria

An extract of existing selection criteria is included below. Filter options that are self-explanatory are not listed here.

Area

Selection list of configured areas of in-production inspection, goods receipt and goods issue. The following areas are available by default:

In-production inspection: Production

Goods receipt: Goods receipt

Goods issue: Goods issue

Initial sample: Initial sample inspection

Gage/test equipment management: Test equipment management

Active

The inspection plan list can be limited to active inspection plans by activating the check box. If this checkbox is not checked, only inspection plans in "Processing" and "Released" status are indicated. The third status of this check box (grayed out) shows all inspection plans. This status is set initially.

Article number

Filtering for article number with regard to article-related inspection plans.

Article designation

Filtering for article designation/name with regard to article-related inspection plans.

Article group

If filtering according to an article group inspection plan is requested, the symbol  can be used to open the article group tree. The Accept and Cancel functions are available here.

Operation

Operation number

Customer number

Direct entry or opening of the customer catalog with adoption of the number of the entry selected there.

Supplier number

Direct entry or opening of the supplier catalog with adoption of the number of the entry selected there.

Manufacturer number

Direct entry or opening of the manufacturer catalog with adoption of the number of the entry selected there.

Field descriptions***Inspection plan header*****Area, inspection plan number, inspection plan index**

"Area", "Inspection plan number" and "Inspection plan index" ensure unambiguity across all existing inspection plans. The area can be selected. The inspection plan number and the inspection plan index can be entered alphanumerically. All 3 fields are mandatory.

The entry of the three pieces of information must be unambiguous, i.e. there must be no inspection plan which already contains this information. It is possible to add specific information by allocating a structured inspection plan number. This may be useful for subsequent sorting. Alternatively, the inspection plan number can also be set to be the same as the article or gage/test equipment group number. The inspection plan index corresponds to the inspection plan version. If an existing inspection plan version is to be changed, but a mere change is not possible since this inspection plan was already used when the inspection order was created, the original inspection plan has to be copied and the inspection plan index has to be modified (e.g. increased by one). With regard to subsequent evaluations (e.g. presentation of cross-order control charts with different underlying inspection plan versions but identical inspection plan number), the inspection plan number should be retained if possible.



As regards the creation of calibration inspection plans, only 24 characters may be entered in the inspection plan number field. Depending on the system configuration, the inspection plan number is entered in the activity calendar in the field "Project number". The field "Project number" is limited to 24 characters there.

Inspection plan type

A distinction is made between the inspection plan types: "Article inspection plan" and "Group inspection plan". The type "Article inspection plan" allows for allocating an article, for which this inspection plan is to be created, from the article master data. As regards the creation of an article group inspection plan, the option "Group inspection plan" is to be selected. This allows for selecting an article group. The selected article group is indicated in a separate field in the overall tree structure.

Article

Entry of article number. If known, the article number may be entered directly. Otherwise, the article catalog can be opened, the searched article can be identified and accepted using the filter and sorting criteria. When selecting an article, the drawing issue number, article designation/name, customer article number and drawing number are transferred from the master data record and indicated in the relevant fields.



If the field "article number" or "article group" is shown in gage/test equipment management, no entries must be made here.

Drawing issue number

The drawing issue number may be entered directly as the article number.



If the article number and the drawing issue number are entered directly, the master data record is identified based on this information when the entry is saved, and subsequently the article designation/name, the customer article number and the drawing number are indicated.



The combination of article and drawing issue number identifies the article unambiguously. The definition and use of a drawing issue number is not mandatory.



If the field Drawing issue number is shown in gage/test equipment management, no entries must be made here.

Operation assignment

One of the two assignment types "One inspection plan for each operation" and "One inspection plan for all operations" can be selected.

The operation assignment defines whether the characteristics of various operations ("One inspection plan for all operations") shall be included in the inspection plan to be generated, or whether the inspection plan shall only include the characteristics of one operation. With the setting "One inspection plan for each operation", the fields "Operation" and "Operation Designation" are shown. Otherwise, the operation information is assigned in the area of the inspection plan characteristics. Wherever possible, the setting "One inspection plan for all operations" should be used. For example, this allows for printing all inspection characteristics of an item even if it comprises several operations.



In areas with no assignment to a production order, the option "One inspection plan for all operations" should be used. The advantage of this is that the "fictional" operation must be entered in the inspection plan header only once instead of an entry for each inspection plan characteristic. The prerequisite for this inspection plan variant is the corresponding system configuration which has to be performed by MPDV Consulting.

Operation, operation designation/name

If operations are used, an entry must be made in only one of the two fields "Operation" or "Operation Designation". If an inspection requirement/inspection order shall automatically be created by the logon of a production order, only the operation number shall be entered in the field "operation".

It is to be observed that these fields are used as search criteria when inspection orders are generated subsequently. This means that if only the designation is entered for the operation in the inspection plan, this precise information must also be included for the generation of inspection orders.



Even in areas with no assignment to a production order (e.g. in goods receipt and in gage/test equipment management), a "fictional" operation is to be assigned (e.g. 9999). This is necessary if you want to generate inspection steps later.

Released / Active

This identifies whether the inspection plan has been released or also has the 'active' status. If the inspection plan has been released and/or is active, the relevant check box is activated. The release and activation of an inspection plan, i.e. its status change, is only possible via the relevant functions in the toolbar. Before activation, a release is required.



Only active inspection plans will be considered in the subsequent inspection order creation.

Released by / on

Shows the HYDRA user who released the inspection plan. In addition, the release date is indicated.

Cavity assignment

The authorization "ipl_cav" is required to show these fields.

A selection between "None" and "Sample" is possible with regard to cavity assignment.



If this field is not available, you need a new program version of this application.

Valid from / to

If required, a validity period may be indicated here instead of "global" activation via the toolbar. This period is considered when the inspection order is created. Since the user will not have a distinct overview of the valid inspection plans in case of activation for a period, the variant of "global" and time-independent activation via the toolbar is preferable. In case of activation via the toolbar, the system precisely monitors whether an active inspection plan already exists for the defined article with the same drawing issue number, customer and supplier indications, if any. If this is the case, the previously active inspection plan is automatically deactivated upon activation.

Inspection type

Characteristic or piece-related inspection

As regards characteristic-related inspection, the relevant characteristic is fully inspected first, if e.g. a total of 5 samples is inspected. As regards piece-related inspection, the characteristic is changed after each value measurement, since one piece is inspected completely in each case.

Action

Create inspection step or Create and immediately release inspection step.

Only released inspection steps can be inspected.

Customer / Supplier / Manufacturer

If a customer/manufacturer or supplier is selected, the relevant inspection plan only applies to the selected company. The customer / manufacturer / supplier is hence the key field of an active inspection plan. In this case, the inspection requirement must also include a company entry so that this inspection plan is used for creating the inspection requirement.

Dynamic modification type

Characteristic or batch-related or none

The authorization "iriscp.dynamic" is required to show these fields.

batch-related: In this case, the transitional definition to be used for dynamic modification is identified on the inspection plan header level. In addition, the initial inspection severity is defined on the inspection plan header level. Finally, the reference for the dynamic modification standard is required on the inspection plan characteristic level. The selection, however, is limited to the dynamic modification standards referencing to the same inspection severity definition as the allocated transitional definition.

characteristic-related: In this case, the transitional definition, the initial inspection severity and the dynamic modification standard are referenced on the inspection plan characteristic level. However, the inspection severity definition of the assigned transitional definition and the assigned dynamic modification standard have to match.

none: The function of dynamic modification is not activated for this inspection plan.

Transitional definition

The authorization "iriscp.dynamic" is required to show these fields.

This field is only available if a batch or characteristic-related dynamic modification was selected before. The available inspection severities are provided and the change between inspection severities is controlled.

Initial inspection severity

The authorization "iriscp.dynamic" is required to show these fields.

This field is only available if a batch or characteristic-related dynamic modification was selected before. It defines the inspection severity applied to the first goods receipt in accordance with the principles for the dynamic modification history.

EMU form

This can be used to select the form types available for initial sampling. At present, only the form type "VDA volume 2, 4th edition" is supported. According to the form type, an appropriate list of entry categories is shown in the inspection plan characteristics.

Editing functions

The key fields "Area", "Inspection plan" and "Inspection plan index" cannot be modified in the editing mode.

Toolbar



Copy

The following dialog opens for copying an inspection plan.

The destination type and area can be entered here. Usually, it should be identical with that of the source inspection plan. Subsequently, the new inspection plan number and index are to be entered. If a new version of an inspection plan is created, the destination plan number is selected identically and only the inspection plan index is modified.



Activate

Function authorization: `iplp.activate`

Switches the inspection plan to the "active" status.



Deactivate

Function authorization: `iplp.deactiv`

Resets the inspection plan from the "active" status to the "released" status.



Release

Function authorization: `iplp.release`

Sets an inspection plan from the "in process" status to the "released" status.

**In process**

Function authorization: iplp.inprocess

Sets an inspection plan from the "released" status to the "in process" status.

Detail application printing

Function authorization	iplp.print
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The print dialog of the inspection plan header opens a list of available reports. These are Word forms. The potential contents of these forms are defined by the Web services available in this context. The quality management master data include definitions of form entries, i.e. the contents of the form list of the relevant printing dialog. This is also where the basis for new forms is created and the relevant form characteristics are defined. Changes in the contents and design of forms require an appropriate license.

Toolbar printing

Apart from the standard functions, no other special functional buttons are available.

Detail application inspection plan characteristics

The detail application of Inspection plan characteristics is almost identical to the application of the characteristic master data. For this reason, only modifications or new features are addressed here.

**Go to**

For more details on characteristics definitions, please refer to the functional description of the document [MOC_CharacteristicsQM](#).

In the inspection plan characteristics level, the relevant inspection plan characteristics are assigned to the previously defined inspection plan header. The assignment is made via a new entry by activating the characteristic catalog and accepting the characteristic highlighted there. By accepting the characteristic, all master data entries are copied into the inspection plan characteristic. Subsequently, any (copied) information can be modified and/or amended. Amendments are frequently made in the characteristic designation in order to define it in more detail.

It is possible to create a characteristic which does not exist in the characteristic catalog. This, however, is only recommended in exceptional cases, since all analyses (e.g. the failure mode analysis) are based on characteristics from the characteristic catalog. For this reason, thorough maintenance of the characteristic catalog is advisable.

Prior to amending specific characteristic data, various properties and settings may be defined.

Field descriptions

Inspection plan characteristics

AFO

The sequence of the subsequent inspection is determined by the work sequence number (AFO). The entry must be unambiguous. Ideally, the new entry should define the work sequence number in increments of ten. This allows for inserting a new characteristic between two existing ones at a later date.

Characteristic number

Number of the characteristic selected from the master data.

Characteristic type

This is where the decision as to whether the subsequent inspection is performed by collecting measured values (variable) or by indication of the number of failures detected (attributive) is made. As regards the attributive inspection, the decision is frequently only based on Pass and/or Fail. Other types of characteristics are the failure collection chart and the information characteristic. The information characteristic is only used to display a document during the inspection sequence. According to the input type, appropriate sampling schemes are available at the bottom.

Inspection result base

A distinction is made as to whether all samples or only the last recorded sample is used to determine the inspection result (Pass/Fail).

Mandatory inspection

If this click field is set, one measured value has to be recorded as a minimum requirement, before an inspection order with this characteristic can be completed.

Calculate characteristic

This designates a characteristic to be calculated.

Formula

Details on this are available in the manual of the CAQ characteristic master data.

Initial sample creation

Selection list of creation categories for initial sampling. The contents depend on the form type selected in the inspection plan header. The inspection plan characteristics are summarized according to the entry category in the subsequent initial sample inspection report, and a separate page is printed for each entry category.

Copy characteristic

This can be used to define which characteristic is to be copied from an initial sample inspection plan into a production plan, for instance. Only the characteristics which were marked as relevant in this field are copied.

Details

This is used to define whether the contents of the "Details" tab are to be defined in the inspection plan with regard to this characteristic, or whether they are to be determined from the underlying master data characteristic on the basis of this inspection plan for the subsequent creation of an inspection requirement. Upon activation of the option "From characteristic catalog", the "Details" tab is hidden. The setting "From characteristic catalog" makes sense if it is a characteristic whose entries can be defined identically for several articles (even across article groups). In this case, the entries can be defined centrally in the master data. This significantly reduces the input and subsequent modification effort.

Specifications

This is used to define whether the contents of the "Specification" tab are to be defined in the inspection plan with regard to this characteristic, or whether they are to be determined from the underlying master data characteristic and/or an entry from the specification list on the basis of this inspection plan for the subsequent creation of an inspection requirement. Upon activation of the option "From list" or "From characteristic catalog", the tabs "Specifications", "Chart 1", "Default values chart 1", "Chart 2" and "Default values chart 2" are hidden. It only makes sense to use the setting "From list" in the case of an article group inspection plan. The specifications list is used if the specifications for distinct characteristics vary from article to article in an article group inspection plan.

Not a cavity

The authorization "ipl_cav" is required to show these fields.

This setting can be used to identify a characteristic as not cavity-relevant, although the associated inspection requirement states that the associated characteristics (actually) are to be collected in relation to a cavity.

This check box must always be set for attributive characteristics and failure collection chart characteristics.

Dynamically modified

The authorization "iriscp.dynamic" is required to show these fields.

This field is only available if a batch or characteristic-related dynamic modification was activated in the inspection plan header before. This setting can be used to define that a characteristic is not to be dynamically modified despite the selected dynamic modification in the inspection plan header.

Transitional definition

The authorization "iriscp.dynamic" is required to show these fields.

This field is only available if a batch or characteristic-related dynamic modification was activated in the inspection plan header before. The selectable transitional definition provides the potential inspection severities and controls the change between inspection severities. With regard to batch-related dynamic modification, the transitional definition is assigned in the inspection plan header and is not available on the inspection plan characteristic level in this case.

Initial inspection severity

The authorization "iriscp.dynamic" is required to show these fields.

It defines the inspection severity applied to the first goods receipt in accordance with the principles for the dynamic modification history. Only the inspection severities of the previously selected transitional definition are available. With regard to batch-related dynamic modification, the transitional definition is assigned in the inspection plan header and is not available on the inspection plan characteristic level in this case.

Standard

The authorization "iriscp.dynamic" is required to show these fields.

This field is only available if a batch or characteristic-related dynamic modification was activated in the inspection plan title before, and this characteristic was also marked as relevant to dynamic modification. Assignment of the dynamic modification standard from the master data to be used.

Inspection level

The authorization "iriscp.dynamic" is required to show these fields.

This field is only available if a batch or characteristic-related dynamic modification was activated in the inspection plan header before, and this characteristic was also marked as relevant to dynamic modification. An inspection level can only be selected if the assigned standard complies with DIN ISO 3951 or DIN ISO 2859.

AQL

The authorization "iriscp.dynamic" is required to show these fields.

This field is only available if a batch or characteristic-related dynamic modification was activated in the inspection plan header before, and this characteristic was also marked as relevant to dynamic modification. An AQL value can only be selected if the assigned standard complies with DIN ISO 3951 or DIN ISO 2859.

Method

The authorization "iriscp.dynamic" is required to show these fields.

This field is only available if a batch or characteristic-related dynamic modification was activated in the inspection plan header before, and this characteristic was also marked as relevant to dynamic modification. A method can only be selected if the assigned standard complies with DIN ISO 3951.

Sample group

The authorization "ipl_ipsampling" is required to show these fields.

This field indicates to which sample group this characteristic belongs. The sample group specifies the characteristics belonging to a sample / a sampling characteristic.

Sampling

The authorization "ipl_ipsampling" is required to show these fields.

If a characteristic was assigned to a sample group, this check box determined whether a sample is to be generated by means of this characteristic. In this case, this is a so-called sampling characteristic. This check box is to be activated for sampling characteristics. The sampling characteristic has to be assigned to a sample group.

If a sample is to be inspected on the basis of this characteristic, this check box must be inactive.



If this field is not available, you need a new program version of this application.



Inspection plan characteristics assigned to a sample group must be configured so that the details are retrieved from the inspection plan instead of the characteristic catalog, since the sample group assignment can only be made in the inspection plan.



All sample characteristics of the same sample group to be inspected must "end" in the same inspection step. This means the combination of inspection station (machine, machine group) and operation must be kept identical for the sample characteristics in the inspection plan.

If the inspections for the producing machine (including sampling characteristic) are to be performed in the same operation as the inspections of the samples, the following aspects are to be observed in the characteristic planning for this operation:



- The characteristics, on the basis of which the inspections are recorded for the producing machine, and the sampling characteristic must have a specific inspection station (e.g. MACHINE).
- The characteristics, on the basis of which the inspections are recorded for the samples, must be assigned to another inspection station (e.g. LABORATORY).
- In the associated inspection plan, the configuration "PS + inspection station" must be configured as "One PS for each inspection station".
- The configuration of the terminal to which the producing machine was assigned must be allocated to the inspection station "MACHINE" for the previously described example

in the "QM functions" tab.

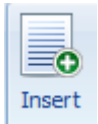
- The configuration of the terminal on which the samples are inspected must be allocated to the inspection station "LABORATORY" for the previously described example in the "QM functions" tab.



Go to

The fields of the relevant tabs correspond to those of the characteristic master data and are to be taken from the functional description of the document [MOC_CharacteristicsQM](#).

Inspection plan characteristics toolbar



Insert

After selecting a characteristic from the master data, all characteristic information of the master data characteristic is transferred to the inspection plan characteristic.



Due to the selection of the master data characteristic, the documents assigned to the master data characteristic are also assigned to the inspection plan characteristic.



Copy

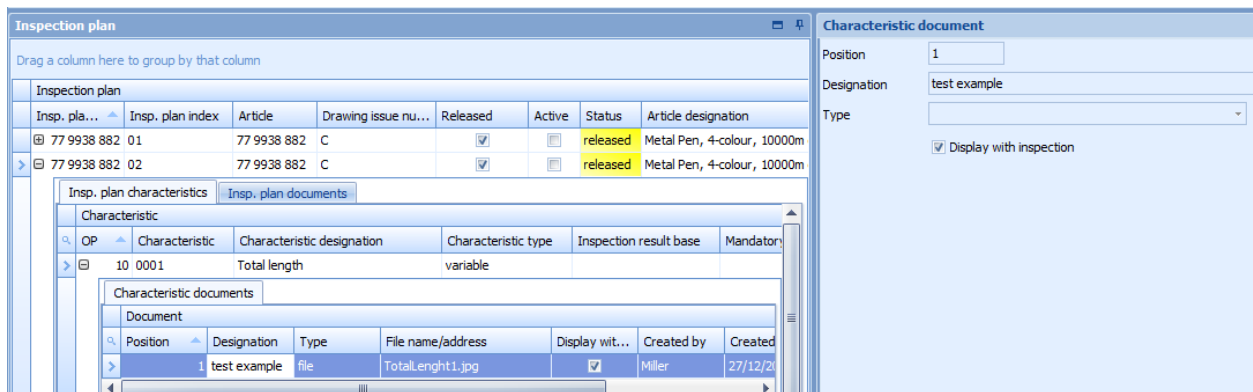
For copying inspection plan characteristics, the data of the highlighted characteristic is opened in the insertion mode. All fields may be edited. For saving, the allocation of an AFO number, which does not yet exist in this inspection plan, is required.



The inspection plan characteristic documents from whose context the 'Copy' button was confirmed are accepted.

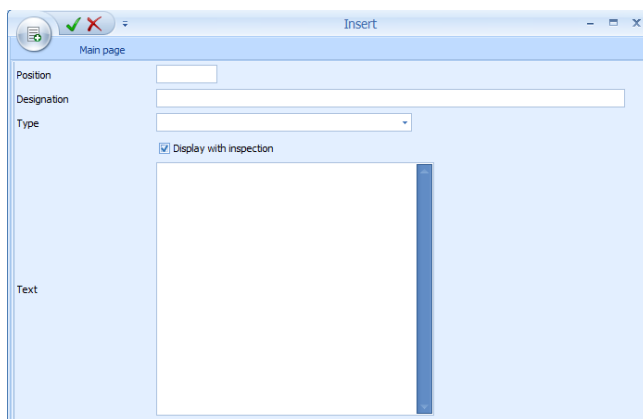
Detail applications "Inspection plan documents" and "Inspection plan characteristic documents"

The above screenshot shows how an inspection plan document is assigned



The above screenshot shows how a document of an inspection plan characteristic is assigned

Upon appropriate activation of the "Inspection plan documents" and/or "Characteristic documents" in the master detail grid, any number of documents may be assigned to each inspection plan and inspection plan characteristic. Upon activation of these tabs, the relevant buttons are activated for maintaining the documents in the toolbar.



As regards document assignment, all formats registered by Windows are available. This means that simple documents (e.g. prepared in Word), drawings of any formats and videos may be assigned. The subsequent display only requires the installation of an appropriate program capable of displaying this format. The documents are opened by the program indicated as link in Windows.

The available file types are "File", "URL" and "Text". As regards the "File" type, the file name including the path may be entered manually. The file type "URL" allows for accessing the Internet and/or Intranet. The third file type, "Text", allows for directly entering a text.

A designation may be assigned to each document entered. In addition, it is possible to define the sequence in which the documents are to be listed. This is effected by using the field Position (numerical entry). The entry within this list must be unambiguous. Apart from this, the check box "Display with inspection" can be used to determine whether the document may be indicated during the inspection process.

Toolbar "Inspection plan documents" and "Inspection plan characteristic documents"

In addition to the standard functions, the button for displaying the documents is available.



Show documents

If a document link is defined, this button opens and shows this document. However, a program, which is able to visualize the linked file type, has to be installed on the PC.